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Latihan 1

ADT

```
constant NMax : integer = 366  
type TabSuhu      : <  Tab : array [1..NMax] of real;  
                        Neff : integer >
```

```
function SearchGtX(T : TabSuhu, X : real) → boolean
```

KAMUS LOKAL

```
check : boolean  
i : integer
```

ALGORITMA

```
check ← false  
i ← 1  
while(i ≤ T.Neff && not(check)) do  
    if(T.Tab[i] = X) then  
        check ← true  
    i ← i+1  
→ check
```

Latihan 2

```
function IdxPengajarMK(T : TabMatkul, P : string) → integer
```

KAMUS LOKAL

```
idx : integer
```

ALGORITMA

```
idx ← 0  
i traversal[1..T.NELMT]  
    if(T>NamaMK = P) then  
        idx ← i  
→ idx
```

Latihan 3a

KAMUS UMUM

```
type TMahasiswa : <  NIM : string;  
                        Nama : string;
```

```

        Nilai : integer { [1..100] }
type TabMhs : < TM : array[1...NMax] of TMahasiswa;
        Neff : integer>

```

```
TMhs : TabMhs
```

```
Temp : integer
```

```
TabCount : array [1..100]
```

```
procedure UrutTabMhs(input/output TMhs : TabMhs)
```

```
KAMUS LOKAL
```

```
    TabCount : array[ValMin...ValMax] of integer
```

```
    i, j : integer {
```

```
    K : integer {indeks array output}
```

```
ALGORITMA
```

```
if(TMhs.Neff > 1) then
```

```
    K ← 0
```

```
    {kalau belum didefinisiin nilai ValMin dan ValMax, bisa ada
    prosedur untuk cari dulu}
```

```
    i traversal[1..NMax]
```

```
        TabCount[i] ← 0 {Biar nilai awalnya ga sampah}
```

```
    i traversal[1..N]
```

```
        TabCount[TMhs.TM.Nilai[i]] ← TabCount[TMhs.TM.Nilai[i]]+1
```

```
{hitung berapa kali angka itu muncul}
```

```
    i traversal[1..NMax]
```

```
        if(not(TabCount[i] = 0)) then
```

```
            j traversal[1..TabCount[i]]
```

```
                K ← K+1
```

```
                TMhs.TM.Nilai[K] ← i
```

Latihan 3b

Karena yang ingin diurutkan adalah nilai, terdapat nilai minimal dan maksimal yang dapat didefinisikan. Oleh karena itu, dapat dihitung pula banyak kemunculan nilai dalam data

Latihan 4

procedure InputTerurut (input/output T : TabInt, input X : integer)

KAMUS LOKAL

i, idx, K : integer

Pass : integer

Temp : integer

found : boolean

ALGORITMA

{Pengurutan array T}

if T.Neff > 1 **then**

 Pass traversal [1..T.Neff]

 K traversal [N..Pass+1]

if(T.Tab[K] < T.Tab[K-1]) **then**

 Temp ← T.Tab[K]

 T.Tab[K] ← T[K-1]

 T.Tab[K-1] ← Temp

{Mencari indeks lokasi X dapat dimasukkan}

i ← 1

while(i ≤ T.Neff **and not**(found)) **do**

if((T.Tab[i-1] < T.Tab[i] **and** T.Tab[i+1] > T.Tab[i]) **or**

(T.Tab[i-1] = T.Tab[i] **and** T.Tab[i+1] = T.Tab[i])) **then**

 idx ← i

 found ← true

 i ← i+1

i traversal[T.Neff+1..idx+1]

 T.Tab[i] ← T.Tab[i-1]

T.Tab[idx] ← X